Exp-2

Program-A

Bit Masking & Logical operations

Given;

- X = A7H = 1010 0111
- Y = 3CH = 0011 1100
- W = 1234H

i) AND operation

X = 1010 0111

Y = 0011 1100

X & Y = 0010 0100 = 24H

ii) OR operation

X = 1010 0111

Y = 0011 1100

X | Y = 1011 1111 = BFH

iii) XOR operation

X = 1010 0111

Y = 0011 1100

X ^ Y = 1001 1011 = 9BH

iv) One's complement of X

X = 1010 0111 = A7H

NOT X = 0101 1000 = 5BH

v) Two's complement of X

One's complement = 5BH

+ 01H

Two's complement = 59H.

Program - B = Bit Setting, Clearing, Shifting & Rotating

Given

D = CAH = 1100 1010

lol = 9234H

1. Set bits 7 and 0

CA = 1100 1010

81 = 1000 0001

1100 1011 = CBH

2. clear bits 3 and 1

CA = 1100 1010

F5 = 1111 0101

1100 0000 = C0H

3. Toggle the lower nibble

CA = 1100 1010

0F = 0000 1111

1100 0101 = C5H

4. Test bit 6

CA = 1100 1010

40 = 0100 0000

0100 0000 = 40H